Madeline LeBreton

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Education

McMaster University – **Bachelor of Engineering in Mechatronics** Sep 2022 - Apr 2027 (Expected)
GPA 3.9/4.0. Engineering Future Leader Award. Dean's Award of Academic Excellence. Dean's Honour List.

Relevant courses: Programming for Mechatronics, Analog and Digital Circuits, Software Development, Probability and Statistics, Scientific Computation, Signals and Systems Embedded Systems Design I and II, Dynamic Models and Control.

Université Grenoble Alpes – **PhITEM** January - April 2025
Exchange Term. Grenoble, France. McMaster Global Learning Ambassador Award.

Publications

1. N. Noor, C. Argentino, A. Irannezhad, ..., **M. LeBreton**, ..., A. Bagger*, D. Higgins*. "Decoupling the Impact of Electronic Structure and Electrode Wettability of Functionalized Iron Phthalocyanine Catalysts for Electrochemical Nitrate Reduction to Ammonia." *J. Am. Chem. Soc.* 2026, 148 (15), 15859–15872. doi:10.1021/jacs.5c22794
   My contribution: phthalocyanine H-cell experiments, NMR and UV-vis quantification, Faradaic efficiencies.
2. N. Noor, U. Nwosu, **M. LeBreton**, A. Schouten, ..., S. Siahrostami*, D. Higgins*. "Elucidating the Impact of the Metal Center Identity and Macrocycle Structure of Molecular Catalysts for Electrochemical Reduction of Nitrate to Ammonia." *ACS Catal.* 2026, 16 (2), 1152–1162. doi:10.1021/acscatal.5c06580
   My contribution: Nitrate reduction work, literature research, created scientific figures and the graphical abstract.
3. T. M. Gibson, **M. LeBreton**, A. Wang, A. Rakhsha, R. Black*, D. Higgins*. "Selected-Ion Flow-Tube Mass Spectrometry (SIFT-MS) for Real-Time Quantitative Measurement of Electrochemical Carbon Dioxide Reduction Reaction Products." *Anal. Chem.* 2025, 97 (43), 24136–24143. doi:10.1021/acs.analchem.5c04792
   My contribution: copper nanoparticle case study verifying the methodology; improved the calculation automation.

Research Experience

Trainee – *European Molecular Biology Laboratory Instrumentation Team* May 2025 - Aug 2025
European Photon & Neutron Science Campus, Grenoble, France | Supervisor: Gergely Papp

- Characterized a motorized laser injection setup for synchrotron crystallography (OXXIUS lasers from 375 to 638 nm, Standa motorized gimbal)
- Wrote Python scan routines that interface a DataRay beam profiler, a power meter and the gimbal to take repeated measurements across laser power and wavelength, automating transmission, spot size and ellipticity mapping across the gimbal range (225 positions per OXXIUS laser)
- Showed that the gimbal repeats a position within ±6 µm and crosses its full range in 1.3 s at maximum speed settings
- Traced lower transmission, ellipticity and spot size in the bottom-left corner of the gimbal range to a misalignment in the mechanical CAD, and proposed the implemented solution

Undergraduate Research Assistant – *National Research Council Canada* May 2024 - Aug 2024
Advanced Materials Research Facility, Mississauga, ON | Supervisor: Dr. Robert Black | Canadian Natural Sciences and Engineering Research Council (NSERC) Undergraduate Student Research Award

- Designed and ran a case study on 25 nm copper nanoparticle catalysts to verify a SIFT-MS methodology for real-time quantification of electrochemical CO_2 reduction products in a flow cell
- Processed over 10,000 real-time data points across ten products and four applied potentials, and improved the Excel-based automation that converts signals into concentrations, production rates, Faradaic efficiencies and partial current densities, including signal deconvolution

Undergraduate Research Assistant – *Higgins Lab, McMaster University* Sep 2023 - Dec 2023
Higgins Lab | Supervisor: Drew Higgins

- Carried out H-cell nitrate-to-ammonia electrolysis experiments and literature research on phthalocyanine catalysts
- Quantified products by NMR and UV-vis spectroscopy and calculated Faradaic efficiencies
- Built 3D molecular models using Python scripting in Blender for the *ACS Catalysis* journal cover

Engineering and Design Experience

Capstone Project – *McMaster University* Apr 2026 - Present
Extended Degrees of Freedom Robotic Visual System with Computer Vision Control, with the Pediatric Image-Guided Therapy Innovation Lab (PCIGITI), The Hospital for Sick Children | Team of 6 | Lab contact: Thomas Looi
- Developing a computer tracking system for surgical teleoperation using the Unitree G1 humanoid
- Working on mechanical design, Simulink modelling and controller tuning, vision, and ROS 2 integration
- Extends the neck of the TWIST2 humanoid teleoperation system with added degrees of freedom

Engineering Intern – *Fundación RIA (Casa RIA)* May 2026 - Aug 2026
Santiago de Compostela, Spain
- Built an R (Shiny, Leaflet) tool that maps walking, cycling and driving isochrones and nearby amenities from OpenRouteService and OpenStreetMap data for architects without GIS training, with QGIS-ready export
- Built an R and Python pipeline for AEMET climate records with trend plots, Excel export and a user application.
- Developed a Kumu systems map of Galicia's water governance from a literature review, as a decision-support tool for exhibition planning

Payload Team Member – *McMaster Rocketry* Sep 2025 - Apr 2026
Experiment and battery housing design, high-power rocket Osiris *(Launch Canada 2026)*
- Won the Payload Challenge at Launch Canada 2026
- Designed a self-healing polymer experiment; compared encapsulation methods and fracture geometries to select urea-formaldehyde microcapsules and a tapered double cantilever beam sample
- Validated the experiment module before prototyping using mass estimates and finite element analysis
- Designed a modular enclosure for four 14500 batteries in Onshape within a −7.5 to 28.5 g acceleration requirement; 3D-printed within tolerances and wrote a CAD guide for the team

Presentations

Oral: EMBL Instrumentation Team, Grenoble, to researchers and external industry partners (2025); Higgins Lab, McMaster University (2023). **Poster:** SIFT-MS copper nanoparticle study, National Research Council Canada (2024).

Awards and Scholarships

Hazelview Engineering Scholarship Sep 2026
McMaster Engineering Future Leader Award Apr 2026
CNL Innovation and Academic Excellence Scholarship Sep 2025
McMaster Global Learning Ambassador Award Jan 2025
NSERC Undergraduate Student Research Award Aug 2024
Ontario Professional Engineers Foundation Undergraduate Scholarship Jul 2023
Entrance awards: Engineering Dean's Award of Excellence, McMaster Award of Academic Excellence, Undergraduate Student Research Award Sep 2022

Leadership and Service

Global Learning Ambassador – *McMaster University* Sep 2025 - Present
- Mentor incoming exchange students and organize orientation events

Mentor to first-year engineering students – *McMaster Engineering Society* Sep 2024 - Present

Ambassador – *Youth Alpine Interrail Network (YOALIN), Switzerland* Jan 2025 - Oct 2025
- Attended summit event in Valposchiavo; discussions on climate, mobility and youth advocacy in Alpine regions

Athlete and Mentor – *Canadian National Cycling Institute* Nov 2021 - Dec 2024
- Represented Canada at the Junior World Championships for Track Cycling; led clinics for women in cycling

Skills

Programming and embedded: Python, C, Java, R, MATLAB, Verilog, Simulink, PIC16F887, FRDM-K64F, FPGA
Instrumentation and laboratory: DataRay beam profiler, optical power meter, Standa gimbal, SIFT-MS, H-cell and flow-cell electrochemistry, NMR, UV-vis spectroscopy
CAD, simulation and data: Onshape, Autodesk Inventor, Blender, QGIS, R
Languages: English (native), French (B2), Spanish (A2)